Dipayan Sarkar

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Research Summary

PhD researcher in bioinformatics and computational biology developing deep learning methods for protein sequence modeling. Work spans sequence-based protein-protein interaction prediction using attention-based architectures with transfer learning from pretrained protein language models, generative modeling of protein binders through autoregressive and diffusion-based approaches. Also builds web platforms that make trained models usable for inference and interaction-network visualization.

Education

Expected Submission 2027 — **Ph.D. in Bioinformatics**, *University of North Bengal, Department of Bioinformatics.*, India
Supervisor: Dr. Chiranjib Sarkar, Assistant Professor, Department of Bioinformatics.
Focus: Framework Development for Predicting Protein-Protein Interaction Networks Using Deep Learning Algorithms on Protein Sequences.

2021 — **M.Sc. in Botany**, *University of North Bengal, Department of Botany.*, India, 77.25% - First Class
Specialization: Biochemistry.
Dissertation: “A review on the responses of aquatic macrophytes to phenolic compounds with special reference to their tolerance strategies and future implications.”
Supervisor: Dr. Swarnendu Roy, Assistant Professor, Department of Botany.

2019 — **B.Sc. (Hons.) in Botany**, *Ananda Chandra College, University of North Bengal*, India, 63.50% - First Class

Research Experience

2022–Present — **PhD Research Scholar**, *Department of Bioinformatics, University of North Bengal*, Siliguri, India
Department of Bioinformatics
- Deep learning architectures for sequence-based protein-protein interaction prediction using Attention-based hybrid architecture with transfer learning.
- Sequence based generative modeling of protein binder generation using autoregressive approaches.
- Development of web-based platforms for model inference and protein interaction network visualization

Publications

2025 — **Sarkar, D.**, & Sarkar, C. **AttnSeq-PPI**: Enhancing protein-protein interaction network prediction using transfer learning-driven hybrid attention. ***Biochimica et Biophysica Acta (BBA)—Proteins and Proteomics***, **1874**(1), 141102. doi:10.1016/j.bbapap.2025.141102

2026 — **Sarkar, D.**, & Sarkar, C. **ARACoFusion**: Uncertainty-aware calibrated deep learning for protein-protein interaction network prediction in *Arabidopsis thaliana*. **bioRxiv** preprint. doi:10.64898/2026.05.22.727120

2026 — **Sarkar, D.**, & Sarkar, C. **MoE-Bind**: Guiding De Novo Protein Binder Generation with Sparse Experts. **bioRxiv** preprint. doi:10.64898/2026.06.13.732043

2026 **Sarkar, D.**, Bardhan, K., & Sarkar, C. **Deep-Interact Studio**: An Interactive Deep Learning Model Building Platform for Biomolecular Interaction Prediction. **bioRxiv** preprint. doi:10.64898/2026.07.02.736034

Research Software Developed

Web Servers

Deep-Interact Studio Interactive web platform for building, training, and comparing custom deep learning models for protein–protein, drug–target, RNA–protein, and protein–DNA interaction prediction, with integrated interpretability.
`deepinteract.compbiosysnbu.in`

AttnSeq-PPI Deep learning-based protein–protein interaction prediction platform.
`attnseqppi.compbiosysnbu.in`

Packages

EGRNi R package for ensemble gene regulatory network inference, combining multiple inference methods to improve reconstruction accuracy of gene regulatory networks.
Sarkar, C., **Sarkar, D.**, Parsad, R., & Mishra, D. *CRAN*, v0.1.6
`cran.r-project.org/package=EGRNi`

Skills

- **Programming**: Python (PyTorch), R, C
- **Deep Learning**: Transformers, attention mechanisms, transfer learning, pre-training and finetuning.
- **Generative AI**: Autoregressive generative models for de novo protein and protein binder design.
- **Scalable Training**: GPU-based training, parallel training on HPC environments
- **Bioinformatics**: Biological sequence analysis, protein-protein interaction and biomolecular network prediction.
- **Scientific Computing**: NumPy, Pandas, Matplotlib, Seaborn.
- **Deployment**: Nginx, Docker, Flask, FastAPI, Linux server deployment, model-backed web applications.
- **Tooling**: Git, GitHub Actions (CI), `uv`, Conda.

Talks & Presentations

2025 **A Webtool for Transfer Learning Based Protein-Protein Interaction Network Prediction Using Hybrid Attention** - Oral presentation, *Anusandhan 2025* (SARANSH, Wiley-sponsored), Bose Institute, Kolkata, India, 7 March 2025.

Awards

2021 **CSIR-UGC NET with JRF**, Life Sciences - June 2021
All India Rank 216 | Junior Research Fellowship, CSIR-HRDG, Government of India

2022 Qualified **GATE** (Graduate Aptitude Test in Engineering) - Life Sciences (XL)

Grants & Computing Resources

2025 **IndiaAI Compute Initiative** - Awarded GPU cloud compute (NVIDIA L4 GPU) under the IndiaAI Mission for the project *Deep learning-based framework for protein-protein interaction network prediction* (Project ID: P1-S2025070964, Sep 2025 - Aug 2026).

2025 **Google TRC (TPU Research Cloud) Award** - Awarded access to GCP - TPUs (v4, v5e, v6e) for machine learning research.